

Hospital Resource Optimization: an investigation into the capacity and utility needs of hospitals and their respective resources.

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Table of Contents

<u>INTRODUCTION</u>	<u>3</u>
<u>INTRODUCTION OF VARIABLES AND MODEL</u>	<u>4</u>
DEVELOPMENT AND MANIPULATION OF THE MODEL	5
<u>ALGORITHM AND CODE</u>	<u>6</u>
<u>ANALYSIS (MANU, LUCAS, GABRIEL)</u>	<u>9</u>
TABLE A- INPUT PARAMETERS ACROSS ALL 12 HOSPITAL TYPES	10
SCENARIO ANALYSIS – RATIONALE, CONSTRAINTS, BOTTLENECKS	11
SUMMARY	13
<u>AREAS FOR IMPROVEMENT</u>	<u>14</u>
AREAS TO IMPROVE MATHEMATICAL MODELLING	14
AREAS TO IMPROVE THE ALGORITHM	14
<u>CONCLUSION</u>	<u>15</u>
<u>SOURCES</u>	<u>16</u>

Introduction

Hospitals

Introduction of variables and model

To model the theoretical patient capacity at a hospital, we have elected to use the following variables:

$d = \text{number of doctors}$

$n = \text{number of nurses}$

$e = \text{number of equipment * (bundled)}$

$b = \text{number of beds}$

$p = \text{number of patients}$

Since hospitals operate continuously (24/7), staffing variables must account for 3 daily shifts:

$d_p = \frac{3}{p_d}$, where p_d is patients per doctor per shift

$n_p = \frac{3}{p_n}$, where p_n is patients per nurse per shift

Based on these variables, we have developed the following constraints:

$$\text{Base constraint: } P \leq \left(\frac{d}{d_p}, \frac{n}{n_p}, \frac{e}{e_p}, \frac{b}{b_p} \right)$$

where x_p refers to the quantity of variables per patient

$$\text{Budget Constraint: } C_d \cdot d + C_n \cdot n + C_e \cdot e + C_b \cdot b \leq B$$

where C_x refers to the cost of variable x and B refers to overall hospital budget

$$\text{Doctor Minimum Staffing Constraint: } d \geq d_{\min}$$

$$\text{Nurse Minimum Staffing Constraint: } n \geq n_{\min}$$

where $d_{\min}$ and $n_{\min}$ represent regulatory minimum staffing requirements

*: Bundled equipment refers to a combination of equipment used in hospitals, such as, monitors, ventilators, imaging systems...

$$\text{Area Constraint: } A_e \cdot e + A_b \cdot b \leq A_T$$

where A_x refers to the area taken by variable x and A_T refers to total area available

$$\text{Non-negativity: } (d, n, e, b, p) \geq 0, \quad \text{and} \quad d, n, e, b \in \mathbb{Z}$$

Development and manipulation of the model

$$f = \max(p)$$

Since p is being maximized, we can re-write the base constraints in the form:

$$d = d_p \cdot p, \quad n = n_p \cdot p, \quad e = e_p \cdot p, \quad b = b_p \cdot p$$

We can substitute these values into the budget constraint:

$$C_d \cdot (d_p \cdot p) + C_n \cdot (n_p \cdot p) + C_e \cdot (e_p \cdot p) + C_b \cdot (b_p \cdot p) \leq B$$

We can re-arrange the equation to isolate p :

$$p(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p) \leq B$$

$$p \leq \frac{B}{(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p)}$$

We can repeat the steps to write the area constraint in terms of p :

$$A_e \cdot (e_p \cdot p) + A_b \cdot (b_p \cdot p) \leq A_T$$

$$p(A_e \cdot e_p + A_b \cdot b_p) \leq A_T$$

$$p \leq \frac{A_T}{(A_e \cdot e_p + A_b \cdot b_p)}$$

Hence, we can write our model as:

$$p \leq \min \left(\frac{B}{(C_d \cdot d_p + C_n \cdot n_p + C_e \cdot e_p + C_b \cdot b_p)}, \frac{A_T}{(A_e \cdot e_p + A_b \cdot b_p)} \right)$$

With the minimum staffing constraints:

$$d \geq d_{\min}$$

$$n \geq n_{\min}$$

Which, using the linear relations above, can be re-written as:

$$p \geq \frac{d_{\min}}{d_p}$$

$$p \geq \frac{n_{\min}}{n_p}$$

Therefore, the final range and constraint for p is given as:

$$\max\left(\frac{d_{\min}}{d_p}, \frac{n_{\min}}{n_p}\right) \leq p \leq \min\left(\frac{B}{C_d d_p + C_n n_p + C_e e_p + C_b b_p}, \frac{A_T}{A_e e_p + A_b b_p}\right)$$

Alongside the non-negativity and integer constraints:

$$(d, n, e, b, p) \geq 0, d, n, e, b, p \in \mathbb{Z}$$

This model, with its related constraints, can be developed into a model on Python that can solve for the maximum capacity a hospital can maintain with a set of given variables.

Algorithm and Code

We translated the mathematical model into a functional and accessible tool by developing a full-stack web application named *IEcare*.

[Link to IEcare](#)

(For the development of the UI and UX features we used “Claude Code” by Anthropic to support the developing process.)

We built the application using Next.js (a React based framework) with TypeScript, which was chosen as it reduces the likelihood of runtime errors when handling the scale of numerical values used in this model. The core optimization engine is based on the javascript-lp-solver library, which is a lightweight implementation of the Simplex method that solves linear programming problems directly in the browser or on a local server. By using this configuration, we avoided requiring users to install any software so that users can simply open the URL and calculate the optimal resource allocations instantly. The application also supports direct exporting of results to an excel file to make the result analysis easier and more intuitive.

The main challenge was to translate the mathematical model, that was developed in the preceding section, into a format the LP solver can work with. We implemented this by creating a constraint matrix, where each row represented a specific constraint, and each decision variable specifies its coefficient in every constraint it participates in. For example, the doctor capacity constraint which stated the number of doctors needed and must not exceed the number of doctors hired, is encoded by assigning ' p ' a coefficient of $\frac{3}{p_d}$ in the *doctor_cap* row, and assigning d (number of doctors employed), a coefficient of -1 in the same row with the constraint bounded at 0.

This formulation lets the solver determine the optimal patient count and the resource levels at the same time. That way, the solver reads it as: $\frac{3}{p_d}$ times however many patients, minus number of doctors must be ≤ 0 . In other words, a way of saying "you need enough doctors for your patients."

A critical detail is the 24/7 shift coverage multiplier. Hospitals operate continuously which requires three 8-hour shifts per day. If one doctor can manage p_d patients during a single shift, then covering one patient around the clock requires $\frac{3}{p_d}$ doctor full time equivalents. For instance, if one doctor handles 3 patients per shift ($p_d = 3$) then each patient requires $\frac{3}{3} = 1$ doctor full time equivalent for full day coverage. This multiplier is applied equivalently to nursing staff. Equipment and beds which are physical assets do not rotate with shifts, so they use their ratios directly without the $3x$ factor.

The budget constraint aggregates all resource costs in a linear way which means that each unit of doctor's nurses equipment and beds contributed its respective annual cost to the total which cannot exceed the annual budget constraint, B .

Integer Solution search

By default, the LP solver operates in a continuous space; meaning that it may return fractional solutions. In practice this would be senseless for a hospital, as non-integer values of resources cannot be utilized. Therefore, our implementation included a dedicated integer solution search. First, the continuous LP relaxation is solved to get the theoretical maximum P , which serves as an upper bound for all solutions. The algorithm is then iterated downward from the upper bound and computes the exact integer resources which are required based on the given patient count. The included ceiling function rounds up; for example, if 14 patients would require 7.1 monitoring stations the model rounds up to 8. The algorithm checks two feasibility conditions which are: total cost less than or equal to the budget and area taken is less or equal to than the total available space. The first value that satisfies both conditions is the optimal integer solution and the search terminates, and since the search starts from the highest boundary possible, the first feasible solution found is guaranteed to be optimal.

[Link to flowchart of the algorithm](#)

This approach is computationally high efficient for the scale of our problem, and the gap between the LP optimum and the integer optimum is small, so the loop executes only a handful of iterations. For larger scale problems, with more decision variables, a mixed integer programming solver would be more appropriate.

Sensitivity Analysis Engine

Beyond only finding a single optimal solution, the most critical question for hospital administrators is how robust the result is. That means, what would happen if the budget changed by 10% or what if the constraint input changes by - 20%. The sensitivity analysis module addresses these kinds of questions in a systematic way.

[Link to sensitivity analysis module](#)

The engine performs three main distinct analyses each requiring hundreds of solver evaluations (approximately 1,500 in total). These include:

1. Parameter Sweeps:

Each input parameter is varied individually from -50% to +50% (*ceteris paribus*) of its baseline value in incremental steps. For each variation the solver reoptimizes and produces a curve that visualizes how patient capacity responds to alterations in the given parameter. These curves reveal if relationships are linear, and follow a common trend, or contain sharp transitions when a constraint may become binding.

2. Tornado Diagram

To quickly identify the most important and influential parameters the algorithm evaluates the impact of a -20% and +20% change in each parameter. The results are ranked by sensitivity from the bottom to the top.

3. Approximate Shadow Prices (Marginal Values)

For each major constraint (budget, space, doctor staffing, nurse staffing) the engine computes a finite difference approximation of the shadow price (λ), which is the rate at which the objective function improves per unit change of each constraint.

All sensitivity results can be exported to a multi-sheet excel workbook, providing a complete package which can be analyzed or made available for shareholders in a real-world context.

Analysis (Manu, Lucas, Gabriel)

The model finds P^* , the maximum number of patients a ward can hold concurrently around the clock. The formula is $P^* = \max(P_{\text{min_staff}}, \min(P_{\text{budget}}, P_{\text{space}}))$. Whichever upper bound is tighter determines the ceiling. The table below explains each constraint type and what it means operationally.

Concept	Where it occurs	What it means in practice
Budget binding	8 of the 12 hospitals	The ward spends its entire annual budget before its floor fills up. Every euro directly converts to patient capacity. The shadow price of an extra euro equals 1 divided by the cost per patient slot. Adding more floor space changes nothing until the budget constraint is resolved first.
Space binding	Private Premium, Military, Psychiatric	The floor is physically full before the money runs out. The shadow price of an extra square metre equals 1 divided by the floor area per patient slot. Adding more budget has zero effect on P^* . The correct intervention is expanding the floor, reducing the footprint per bay, or relocating to a larger building.
Dual binding (budget + space)	ICU only	Both constraints saturate at the same patient count — there is no slack in either resource. Both shadow prices are positive. Relaxing only one of them does nothing because the other immediately becomes the new ceiling. This is the most efficient operating point in the model and the only scenario where every euro and every square metre is simultaneously productive.
Min staffing floor (lower bound)	Rural Cottage Hospital	Regulations require a minimum number of doctors and nurses on duty at all times, regardless of how many patients are in the building. This is a lower bound on P^* , not an upper bound. Budget and space both have slack, but the mandatory headcount consumes so much of the budget that P^* cannot be freely optimised. Improvement requires either regulatory change or enough additional budget to fund the floor staff and still have capacity left over.
Non-binding constraint (shadow price = 0)	Space in most budget-bound hospitals	When a constraint has slack it is non-binding, and its shadow price is zero — improving it has no effect on P^* whatsoever. In the eight budget-bound hospitals, space is non-binding. Investing in a building extension before addressing the budget constraint would waste capital entirely. The test is simple: compare P_{budget} (budget $\div$ cost per slot) with P_{space} (space $\div$ space per slot). The smaller number identifies the binding constraint.
Bottleneck within the binding constraint	Staff salaries in budget-bound hospitals	Even within a binding constraint, one component dominates. In budget-bound hospitals, staff salaries running across three daily shifts account for 85–95% of the cost per patient slot. Equipment — monitors and beds — makes up the remainder. This means improving the number of patients each clinician can safely manage per shift has roughly ten times more impact on P^* than halving equipment costs. In space-bound hospitals, bed footprint similarly dominates over monitor footprint.

Why the binding constraint matters

Two hospitals with identical budgets and identical floor space can have completely different binding constraints if their cost structure or room sizes differ. Knowing which constraint is active is the only information needed to choose the right intervention. Investing in floor space when budget is binding, or adding budget when space is binding, produces no improvement at all — the money or the square meters simply go unused.

Table A- Input Parameters Across All 12 Hospital Types

Annual costs in €/yr, fully loaded. Equipment costs are annualised. Staffing ratios are per 8-hour shift — the model multiplies by 3 for 24-hour coverage. Binding constraint is colour-coded: red = budget, amber = space, blue = dual binding, purple = min staffing floor.

Hospital	Budget €M	Space m²	Dr cost €k	Nur cost €k	Mon cost €k	Bed cost €k	Pts/Dr shift	Pts/Nur shift	m²/Bed	LOS days	P*	Binding constraint
Large National Referral	25.0	2,400	130	65	4.0	7.0	6	4	8	6	201.4	Budget
University Teaching Hospital	18.0	1,600	160	70	7.0	9.0	4	3	9	8	87.4	Budget
Regional District General	8.0	800	120	60	5.0	8.0	5	3	8	7	55.5	Budget
Small Community Hospital	3.0	300	100	55	3.0	5.0	6	4	7	5	30.6	Budget
Intensive Care Unit (ICU)	12.0	600	200	80	20.0	15.0	2	1	12	4	20.9	Budget
Paediatric Specialist Hospital	10.0	900	140	75	6.0	8.0	4	2	12	9	43.5	Budget
Private Premium Hospital	20.0	700	180	90	12.0	20.0	4	2	15	5	40.2	Space
Day Surgery Centre	5.0	400	110	58	3.0	4.0	8	5	6	1	60.6	Space
Rural Cottage Hospital	1.2	150	90	50	2.5	4.5	5	3	7	4	10.9	Budget
Military Field Hospital	4.0	200	95	52	1.5	2.5	8	5	5	2	37.3	Space
Psychiatric Inpatient Ward	6.0	1,000	130	65	0.5	3.0	6	2	20	21	36.2	Budget
Pandemic Surge Hospital	30.0	8,000	110	58	8.0	3.0	10	6	4	5	434.8	Budget

Scenario Analysis – Rationale, Constraints, Bottlenecks

Red cells in the budget/space columns indicate a binding constraint ($\geq 95\%$ utilised). The analysis column for each scenario explains why the parameters were set as they were, identifies the binding and non-binding constraints, and names the bottleneck within the binding constraint.

Hospital & type	P*	Annual patients	Budget utilised	Space utilised	Rationale, constraint analysis, and bottleneck
Large National Referral <i>Tertiary / National</i>	201.4	12,249	100%	89%	A flagship government hospital receiving referrals from across the country. Budget and floor space are the largest of any conventional hospital. The mix of senior consultants and registrars allows doctors to carry six patients per shift. Regulatory minimums are high because the hospital must maintain specialist cover across multiple disciplines at all times. Binding: Budget (budget utilised 100%, space utilised 89%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 92% of the €124k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
University Teaching Hospital <i>Tertiary / Academic</i>	87.4	3,988	100%	74%	Jointly funded by the health authority and a university medical school. Senior consultants must supervise trainees during ward rounds, which reduces the number of patients each doctor can actively manage per shift — hence a 4:1 ratio rather than the standard 5:1 or 6:1. Equipment costs are higher because monitoring hardware is also used for research. Nurse cost reflects specialist training at academic level. Binding: Budget (budget utilised 100%, space utilised 74%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 92% of the €206k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Regional District General <i>Secondary / Regional</i>	55.5	2,894	100%	74%	The workhorse of most health systems, serving a city of 300,000–500,000 people. Every parameter sits at the industry benchmark — standard consultant costs, standard staffing ratios, standard bed size. This scenario is the reference point from which all others deviate. Binding: Budget (budget utilised 100%, space utilised 74%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 92% of the €144k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Small Community Hospital <i>Primary / Community</i>	30.6	2,235	100%	85%	Lower-acuity patients mean doctors and nurses can safely supervise more patients per shift. GPs replace hospital consultants, reducing doctor cost to €100k. Equipment is basic. The hospital discharges patients quickly to community care, keeping the average stay to five days. Binding: Budget (budget utilised 100%, space utilised 85%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 93% of the €98k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Intensive Care Unit (ICU) <i>Specialist / ICU</i>	20.9	1,904	100%	56%	Every parameter is driven by the imperative to keep critically ill patients alive moment to moment. One nurse per patient is the international standard for intensive care. Intensivists carry a caseload of two because complex physiological management of a ventilated patient is genuinely full-time work. Monitors cost €20k because they integrate ventilator data, arterial line readings, and continuous waveform display. Each bed bay requires 12 m ² so that a resuscitation team of four or five can work simultaneously at the bedside. Binding: Budget (budget utilised 100%, space utilised 56%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 94% of the €575k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Paediatric Specialist Hospital <i>Specialist / Paediatric</i>	43.5	1,762	100%	74%	Children's hospitals accommodate a parent or guardian alongside the child, which pushes the space requirement per bed to 12 m ² . Paediatric nurses require specialist training and command higher salaries. Doctors supervise fewer patients because children cannot self-report symptoms, making clinical assessment more time-intensive. Stays are longer because paediatric conditions require extended observation before safe discharge. Binding: Budget (budget utilised 100%, space utilised 74%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 95% of the €230k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.

Hospital & type	P*	Annual patients	Budget utilised	Space utilised	Rationale, constraint analysis, and bottleneck
Private Premium Hospital <i>Private / Premium</i>	40.2	2,936	60%	100%	A city-centre location keeps the total floor area to 700 m², but every patient occupies a private en-suite room. Those rooms are large — 15 m² per bed, including a sitting area — which is the highest space-per-patient of all inpatient scenarios. Despite a €20M budget, the floor fills before the money runs out. This is what makes space the binding constraint here rather than budget. Binding: Space (budget utilised 60%, space utilised 100%). Non-binding: Budget (shadow price = 0 — improving these has no effect). Bed footprint accounts for 86% of the 17.4 m² per slot. Reducing bed spacing or using smaller bed formats is the most productive intervention.
Day Surgery Centre <i>Specialist / Day Surgery</i>	60.6	22,121	98%	100%	Patients are admitted and discharged on the same day, which completely transforms the annual throughput calculation. Every concurrent bed turns over 365 times per year. Procedures are standardised and short, so doctors can cycle through eight patients per shift and nurses can manage five. Recovery bays are smaller than inpatient rooms and do not require overnight infrastructure. Binding: Space (budget utilised 98%, space utilised 100%). Non-binding: Budget (shadow price = 0 — improving these has no effect). Bed footprint accounts for 91% of the 6.6 m² per slot. Reducing bed spacing or using smaller bed formats is the most productive intervention.
Rural Cottage Hospital <i>Primary / Rural</i>	10.9	995	100%	60%	The defining feature of this scenario is not its budget or its space — it is the regulatory requirement to maintain at least one doctor and three nurses on duty at all times, regardless of how many patients are in the building. Running that minimum across three shifts costs approximately €420,000 per year, which is 35% of the entire €1.2M budget before a single additional patient slot is funded. Budget and floor space both have slack, but P* cannot fall below the level implied by the mandatory headcount. Binding: Budget (budget utilised 100%, space utilised 60%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 95% of the €110k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Military Field Hospital <i>Specialist / Military</i>	37.3	6,809	65%	100%	A tent or modular structure fixes the available floor area at 200 m² regardless of budget. Medical officers are trained to manage larger caseloads than civilian norms because battlefield medicine protocols prioritise throughput. Equipment is ruggedised and cheap. Field cots require only 5 m² each, but even so the tent fills before the military budget is exhausted — which is why space, not budget, is the binding constraint. Binding: Space (budget utilised 65%, space utilised 100%). Non-binding: Budget (shadow price = 0 — improving these has no effect). Bed footprint accounts for 93% of the 5.4 m² per slot. Reducing bed spacing or using smaller bed formats is the most productive intervention.
Psychiatric Inpatient Ward <i>Specialist / Psychiatric</i>	36.2	629	100%	73%	Psychiatric patients do not require physiological monitoring, so equipment costs are negligible — €500 per year for an occasional vital-signs check. What drives this scenario is space. Regulations in most jurisdictions mandate individual rooms for psychiatric inpatients, and those rooms must also be safe for patients at risk of self-harm. Add proportionate access to shared therapy rooms, dining, and outdoor space and the floor area attributed per patient reaches 20 m² — the highest of any inpatient setting. The ward has 1,000 m² but space still binds before the budget runs out. Binding: Budget (budget utilised 100%, space utilised 73%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 98% of the €166k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Pandemic Surge Hospital <i>Emergency / Surge</i>	434.8	31,739	100%	24%	A converted exhibition hall or sports arena pressed into service during a public health crisis. Emergency decree unlocks a €30M budget and staffing ratios are stretched well beyond normal clinical guidelines — ten patients per doctor per shift, six per nurse. Field cots are laid out at 4 m² each across 8,000 m² of floor. At this scale budget is the binding constraint, and P* is far higher than any conventional hospital. The scenario exists to show what happens when volume is the only objective and clinical quality is secondary. Binding: Budget (budget utilised 100%, space utilised 24%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 90% of the €69k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.

Summary

P* ranges from 3.8 patients in the rural cottage hospital —where regulatory staffing obligations consume most of the budget before any capacity is created — to 376 in the pandemic surge hospital, where both space and budget dwarf anything seen in conventional medicine. Within the realistic band, the national referral hospital reaches 44 patients, and the ICU reaches 7.5.

The biggest lesson across all twelve scenarios is that the binding constraint is institution-specific and must be diagnosed before any investment decision is made. Budget is binding in eight hospitals, driven almost entirely by staff salaries running across three daily shifts. Space is binding in three hospitals where the per-patient floor area is large by design — private rooms, ICU bays, or psychiatric therapy space. The rural hospital stands apart: its problem is neither too little money nor too little space but a regulatory minimum that forces fixed staffing costs regardless of patient volume.

Staffing ratios are by far the most powerful lever available to a budget-bound hospital. Because three shifts of staff must be funded for every concurrent patient, a one-unit improvement in the patients-per-clinician ratio reduces the cost per slot substantially — far more than any plausible reduction in equipment costs. For space-bound hospitals, the equivalent lever is bed footprint: reducing the floor area per bed, whether through architectural redesign or regulatory reclassification, directly unlocks additional capacity without requiring any capital investment in the building envelope.

The overarching principle

Identify which constraint is binding, then attack the largest driver within it. Re-run the model after each intervention because relaxing one constraint shifts the binding point to the next. No single change reaches the maximum — the gains compound when budget, space, and staffing efficiency are all improved in the right sequence.

Areas for improvement

Areas to improve mathematical modelling

The model developed represents the fundamental resources employed in a hospital and provides rudimentary constraints for each resource. In reality, hospitals maintain a much larger number of resources, such as specified instruments available, operation theatres, etc. This requires unique constraints for each resource, meaning that to truly optimize the resources a hospital has available, there would be a large increase in the number of resources and constraints employed. Our model simplified the resources, their respective constraints, and used simple constraints.

Furthermore, our model was developed to be linear in nature, which is often not the case in real life. In reality hospitals often experience economies of scale or diseconomies of scale, meaning that their utilization of resources is not strictly linear. However, we believed that developing a non-linear model would be out of scope for this project, as well as what best suited the requirements of the problem.

Areas to improve the algorithm

Our current implementation relies on the javascript-lp-solver as explained above. While this library is convenient for a browser-based deployment it maintains several notable limitations: it is limited to solving continuous linear programming relaxation and does not feature any built-in support for integer programming. As a result, we implemented our own integer search by iterating downward from the results obtained by the LP solver and then testing each candidate for feasibility. This mutation of the integer LP solver works reliably for our model because the number of constraints and decision variables is small. However, for larger or more complex models this version would become computationally impractical.

In addition, as mentioned above, the current model and solver assume all relationships to be strictly linear. Incorporating nonlinear relationships would require going beyond linear programming entirely toward nonlinear programming or piecewise linear approximations within the Linear Programming framework.

The current algorithm is hard coded for four resource types (doctors, nurses, equipment and beds) with two main constraints (budget and space). A more extensible and applicable architecture would be to allow users to define resource types and constraints by themselves. This would enable the tool to be applied to a wider range of hospital configurations.

Furthermore, the model is static which means that it optimizes for a single point in time with fixed parameters. A natural extension would be to introduce a real time dimension which would allow the model to account for seasonal demand fluctuations, phased capital investments or staff hiring timelines. This would transform the problem from a single period LP into a multi period optimization problem which would significantly increase both its realism but also its computational complexity.

Conclusion

Sources